Excel Cheat Sheet

Formulas, shortcuts, pivot tables and interview answers for Data Analysts

GRIND WITH RITIK

1. MATH AND BASICS

`=SUM(A1:A10)`	Adds the numbers
`=AVERAGE(A1:A10)`	Mean of the numbers
`=MIN(A1:A10) =MAX(A1:A10)`	Smallest / largest value
`=COUNT(A1:A10)`	Counts cells that hold numbers
`=COUNTA(A1:A10)`	Counts non-empty cells
`=COUNTBLANK(A1:A10)`	Counts empty cells
`=ROUND(A1,2)`	Rounds to 2 decimals
`=MOD(10,3)`	Remainder (gives 1)
`=ABS(A1)`	Removes the minus sign

2. LOGIC AND CONDITIONS

`=IF(A1>=50,"Pass","Fail")`	Returns one value if true, another if false
`=IFS(A1>=90,"A",A1>=75,"B",TRUE,"C")`	Several conditions in one formula (Excel 2019 and later)
`=AND(A1>5,B1<10)`	True only if all conditions are true
`=OR(A1>5,B1<10)`	True if any condition is true
`=IFERROR(A1/B1,0)`	Shows 0 instead of an error
`=SUMIF(A:A,"Pune",B:B)`	Adds B where A equals Pune
`=SUMIFS(C:C,A:A,"Pune",B:B,">50")`	Adds C with several conditions
`=COUNTIF(A:A,">=50")`	Counts cells meeting one condition
`=COUNTIFS(A:A,"Pune",B:B,">50")`	Counts with several conditions
`=AVERAGEIF(A:A,"Pune",B:B)`	Average of B where A equals Pune

3. LOOKUP

`=VLOOKUP(E2,A:C,3,FALSE)`	Finds E2 in the first column, returns column 3. FALSE means exact match
`=HLOOKUP(E2,A1:F3,2,FALSE)`	Same idea, but searches across a row
`=INDEX(C:C,MATCH(E2,A:A,0))`	Flexible lookup. Can return values from the left of the search column
`=XLOOKUP(E2,A:A,C:C,"Not found")`	Modern lookup. Looks in any direction (Excel 2021 and 365)
`=MATCH(E2,A:A,0)`	Position of E2 in the column

VLOOKUP cannot look left and breaks if you insert a column. Prefer INDEX/MATCH or XLOOKUP.

4. TEXT

`=LEFT(A1,3) =RIGHT(A1,3)`	First / last 3 characters
`=MID(A1,2,4)`	4 characters starting at position 2
`=LEN(A1)`	Number of characters
`=TRIM(A1)`	Removes extra spaces
`=UPPER(A1) =LOWER(A1) =PROPER(A1)`	Changes the case
`=A1&" "&B1 =CONCAT(A1,B1)`	Joins text
`=TEXTJOIN(", ",TRUE,A1:A5)`	Joins a range with a separator
`=SUBSTITUTE(A1,"-","")`	Replaces text
`=FIND("@",A1)`	Position of a character (case-sensitive)
`=TEXT(A1,"dd-mmm-yyyy")`	Formats a number or date as text

5. DATES

`=TODAY() =NOW()`	Current date / date and time
`=YEAR(A1) =MONTH(A1) =DAY(A1)`	Parts of a date
`=DATE(2026,10,12)`	Builds a date
`=EOMONTH(A1,0)`	Last day of the month
`=EDATE(A1,3)`	Same date 3 months later
`=DATEDIF(A1,B1,"D")`	Days between two dates (use M or Y for months or years)
`=NETWORKDAYS(A1,B1)`	Working days between two dates
`=WEEKDAY(A1)`	Day of the week as a number

6. DYNAMIC ARRAYS (365 / 2021)

`=UNIQUE(A2:A100)`	List of distinct values
`=SORT(A2:B100,2,-1)`	Sorts by column 2, descending
`=FILTER(A2:C100,B2:B100>50)`	Rows that meet a condition
`=SEQUENCE(5)`	Numbers 1 to 5

These spill into the cells below. They do not work in older versions of Excel.

7. PIVOT TABLE IN 6 STEPS

`1 Click inside your data`	Make sure every column has a header
`2 Insert > PivotTable`	Choose a new worksheet
`3 Drag fields`	Rows, Columns, Values and Filters boxes
`4 Value Field Settings`	Switch between Sum, Count and Average
`5 Show Values As`	Percent of total, running total and more
`6 Insert > Slicer`	Click buttons to filter. Use Refresh when the data changes

8. DATA CLEANING TOOLKIT

`Data > Remove Duplicates`	Deletes repeated rows
`Data > Text to Columns`	Splits one column into many
`Flash Fill (Ctrl+E)`	Spots a pattern and fills the rest
`Find and Replace (Ctrl+H)`	Bulk edits
`TRIM and CLEAN`	Remove stray spaces and hidden characters
`Ctrl+G > Special > Blanks`	Selects all empty cells
`Data > Data Validation`	Drop-down lists and input rules
`Power Query (Data > Get Data)`	Import and clean data with repeatable steps

9. KEYBOARD SHORTCUTS (WINDOWS)

`Ctrl+Arrow`	Jump to the edge of the data
`Ctrl+Shift+Arrow`	Select to the edge of the data
`Ctrl+Space / Shift+Space`	Select column / row
`Alt+=`	AutoSum
`F2`	Edit the cell
`F4`	Cycle $ in a reference (A1, A1, A$1, $A1)
`Ctrl+; / Ctrl+Shift+;`	Insert today's date / time
`Ctrl+D / Ctrl+R`	Fill down / fill right
`Ctrl+T`	Turn the range into a Table
`Ctrl+Shift+L`	Turn filters on or off
`Ctrl+1`	Format Cells
`Ctrl+PageUp / PageDown`	Switch sheets

On a Mac, most shortcuts use Cmd instead of Ctrl, but a few differ.

10. CELL REFERENCES

`A1`	Relative. Changes when you copy the formula
`$A$1`	Absolute. Never changes
`$A1 / A$1`	Mixed. Locks the column / the row
`=Sheet2!A1`	Reference to another sheet
`Name Box`	Give a range a name, then use =SUM(Sales)

11. COMMON ERRORS

`#DIV/0!`	Divided by zero. Wrap in IFERROR
`#N/A`	A lookup did not find the value
`#NAME?`	Misspelled function or name
`#REF!`	A referenced cell was deleted
`#VALUE!`	Wrong type of data, for example text in maths
`#####`	Column is too narrow. Widen it
`#SPILL!`	Something blocks the spill range

12. QUICK INTERVIEW ANSWERS

`VLOOKUP vs XLOOKUP?`	VLOOKUP searches the first column and returns from the right. XLOOKUP and INDEX/MATCH look in any direction.
`COUNT vs COUNTA vs COUNTIF?`	COUNT: numbers. COUNTA: non-empty cells. COUNTIF: cells that meet a condition.
`Relative vs absolute reference?`	Relative changes when copied. Absolute (A1) stays fixed.
`What is a pivot table?`	A drag-and-drop summary of big data without writing formulas.
`What is Power Query?`	A tool to import, clean and reshape data with steps you can refresh.
`What is a named range?`	A label for a range that makes formulas easier to read.

13. WHICH CHART TO USE?

`Line`	Trends over time
`Column / Bar`	Compare categories
`Pie`	Parts of a whole (keep it to a few slices)
`Scatter`	Relationship between two numbers
`Histogram`	How values are distributed
`Combo`	Two measures with different scales

14. FORMATTING TIPS

`Ctrl+Shift+1`	Number format with commas
`Ctrl+Shift+4`	Currency format
`Ctrl+Shift+5`	Percent format
`Home > Conditional Formatting`	Colour cells by rules, data bars, icon sets
`View > Freeze Panes`	Keep headers visible while scrolling
`Ctrl+T`	Format as Table: banded rows, filters, auto-expanding ranges